# Complete Empirical Capture of the Agent/Environment Interaction during Natural Behavior across Phylogony, Domain, and Dimension

FreeMoCap Foundation, Inc.

Topic: Scientific Instrumentation for Sensing and Imaging

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## 1. Mission

- **Funnel**: The modern landscape of science is one of loosely connected pockets of hyper specialization within siloed discplines and domains of inquiry.
- ‣ **Gap**: While this approach has led to many advances over the past century, the current model of specialized single-domain advancemnt has failed to create integrative frameworks to combine the individual threads of single-modality advancement into an coherent whole representing a richly interconnected representation nature capable of actionable operation.
- **Hero**: A high quality integrated platform directly designed for the use-case development of complete empirical capture of an awake behaving human-or-non-human-animal (HNHA) would transform the scientific landscape by creating a technical commons whereby specialists and cross-disciplnary researchers can productively interact in serivce of the shared and unified scienfitic envdeavor.

**MISSION STATEMENT** - Build a Platform for **Complete Empirical Capture of the Agent/Environment Interaction** to create Interoperabilty across all Dimensional Domains to align Neuroscience with Robotics

**Complete Empirical Capture:** Complete empirical capture means that we are recording every single possible thing that we can record about a given slice of time over continuous time.
**Agent/Environment Interaction:** Relates to the internal and external state of an actor (human, non-human animal, or artificial system).
**Interoperabilty across all Dimensional Domains:** A "dimensional domain" is any thing that can vary along an orientable dimension. We will a platform that will create seamless interoperability across **species**, **scale**, **time**, **numerosity**, and **complexity**.

## 2. Technology Landscape

Provide analysis of current technology landscape that justifies how the proposed platform technology would accelerate the emergence of new technologies and lines of research. The Technology Landscape Analysis should compare current state-of-the-art technology and should provide a clear description of the team's goal to progress the technology and unlock fundamentally new capabilities. A plot or other figure to quantitatively depict the current technology landscape and desired goal(s) the NSF X-Lab may be included here.

## 3. Outcomes

### 3.1. 7 Year Plan

0. Organize
1. Architect
2. Execute
3. Stabilize
4. Expand
5. Align
6. Establish
7. Maintain

## 4. Senior/Key Personnel Qualifications

Provide a brief description of the qualifications of Senior/Key Personnel on the team. Include technical expertise, experience commercializing technology, past efforts, and other qualifications that will contribute to achieving the team's Mission. Include information for all Senior/Key Personnel, including team members not currently affiliated with the lead organization.

## 5. Team Capabilities Statement

Provide a Team Capability Statement describing how the Senior/Key Personnel on the proposed leadership team bring together complementary expertise in strategic leadership, technical expertise, and/or entrepreneurship, as applicable. The statement should describe how, collectively, the team demonstrates the experience, adaptability, and collaborative approach necessary to execute the Mission. Briefly describe the team's intended governance structure and autonomy during Phase 0 and Phase 1. If applicable, include an overview of networks, partnerships, and capital resources your team currently has and how they will be deployed in support of the proposal.

## Bibliography